EXPERIMENT - 6

Simple Deluge Scripts, Workflows, Testing and Debugging

Loan Application & Pre-Screening Management

Application Name	Loan Application & Pre-Screening Management
Form Name	Loan_Application_Form
Report Name	Loan_Application_Report
Student Name	DAKSHESH N
Student Roll No.	241401017
Experiment	6
Platform	Zoho Creator

Lab Record
Validate loan applications before saving, acknowledge successful submissions, and test the workflow with fictional records and diagnostic messages.

Department of Computer Science and Engineering

Rajalakshmi Engineering College

1. Objective and Workflow Design

To use simple Deluge rules for age, monthly income and consent; handle successful submissions; and inspect errors and logs without sending messages for mock records.

Workflow	Event	Action
Validate_Loan_Application	Created or edited; validations on submission	Reject an applicant under 18, non-positive income or missing consent.
Loan Application Processing	Created; successful submission	For a non-mock email, send an internal review message and acknowledgement. For a missing or example.com email, log a skip message.

2. Implementation Procedure

1. Create the validation workflow for Loan_Application_Form and enter the three simple checks. Use alert with cancel submit so invalid records are not saved.

2. Retain the existing successful-submission workflow. Add a mock-email guard and a short received-for-review acknowledgement.

3. Save the scripts and confirm both workflows are enabled.

4. Run negative tests one at a time, correct the failed input and verify valid submissions in the report.

5. Inspect the info output for mock submissions. Treat receipt as an acknowledgement, not approval.

> **Simple status handling:** the acknowledgement and log state that the application was received for review. No separate status field or approval engine was added. The income threshold is only a retained routing example for manual review, not a lending eligibility decision.

3. Test Conditions

Tests were performed with fictional values only. example.com addresses trigger the no-email branch. No real applicant identity, banking information or real contact destination was used. The date-boundary test uses 06-Oct-2026 as the execution date.

4. Validation Script

Workflow: Validate_Loan_Application. Event: created or edited, validations on submission. The Date of Birth and Monthly Income fields are also mandatory.

```
if(input.Date_Of_Birth > zoho.currentdate.subYear(18))
{
    alert "Applicant must be at least 18 years old.";
    cancel submit;
}
if(input.Monthly_Income <= 0)
{
    alert "Monthly income must be greater than 0.";
    cancel submit;
}
if(input.I_consent_to_a_credit_bureau_check == null ||
   !input.I_consent_to_a_credit_bureau_check.contains("Yes"))
{
    alert "Credit-check consent is required.";
    cancel submit;
}
```

How the Checks Work

The birth date is compared with the date 18 years before today. A later birth date means the applicant is too young. Monthly income must be greater than zero. Consent is a checkbox choice collection, so the script checks both null and whether Yes is selected.

Each failure raises an explanatory alert and cancels the submission. Field-level mandatory checks may run first; an unchecked mandatory consent field can therefore be rejected before the custom consent alert appears.

Debugging principle: test one invalid input at a time so the displayed message can be associated with the correct rule. A successful script save checks syntax; runtime tests verify behaviour.

5. Successful-Submission Script

The following is the saved action logic, formatted for the record. Long message lines wrap for readability. The outer application-definition wrapper is omitted.

```
if(input.Email != null && input.Email != "" &&
   !input.Email.endsWith("@example.com"))
{
    if(input.Monthly_Income >= 50000)
    {
        sendmail
        [
            from :zoho.adminuserid
            to :zoho.adminuserid
            subject :"Loan Application - Initial Review"
            message :"A new loan application has been submitted. The applicant's monthly income is ₹50,000 or above
    and the application qualifies for initial review."
        ]
    }
    else
    {
        sendmail
        [
            from :zoho.adminuserid
            to :zoho.adminuserid
            subject :"Loan Application - Further Review"
            message :"A new loan application has been submitted. The applicant's monthly income is below ₹50,000
    and the application requires further review."
        ]
    }
    sendmail
    [
        from :zoho.adminuserid
        to :input.Email
        subject :"Loan Application Received"
        message :"Your loan application has been received for review. A loan officer will contact you for document
    verification. This acknowledgement is not a loan approval."
    ]
}
else
{
    info "Application received for review. Email skipped for mock data or missing email.";
}
```

Actual email delivery was not exercised. This Creator sendmail workflow is separate from the standalone Zoho Mail connection/API demonstration in Experiment 7.

6. Workflow Configuration Evidence

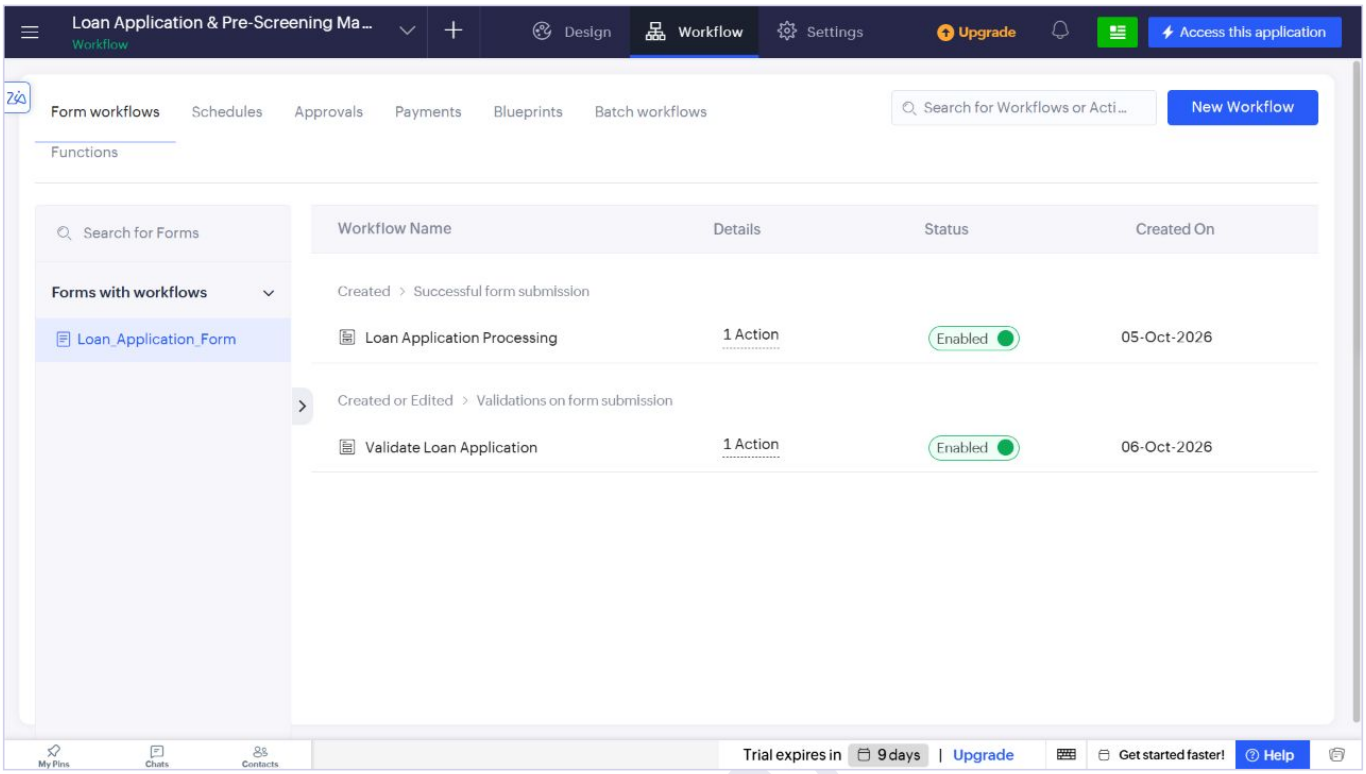


Figure 6.1 - Both submission workflows are present and enabled.

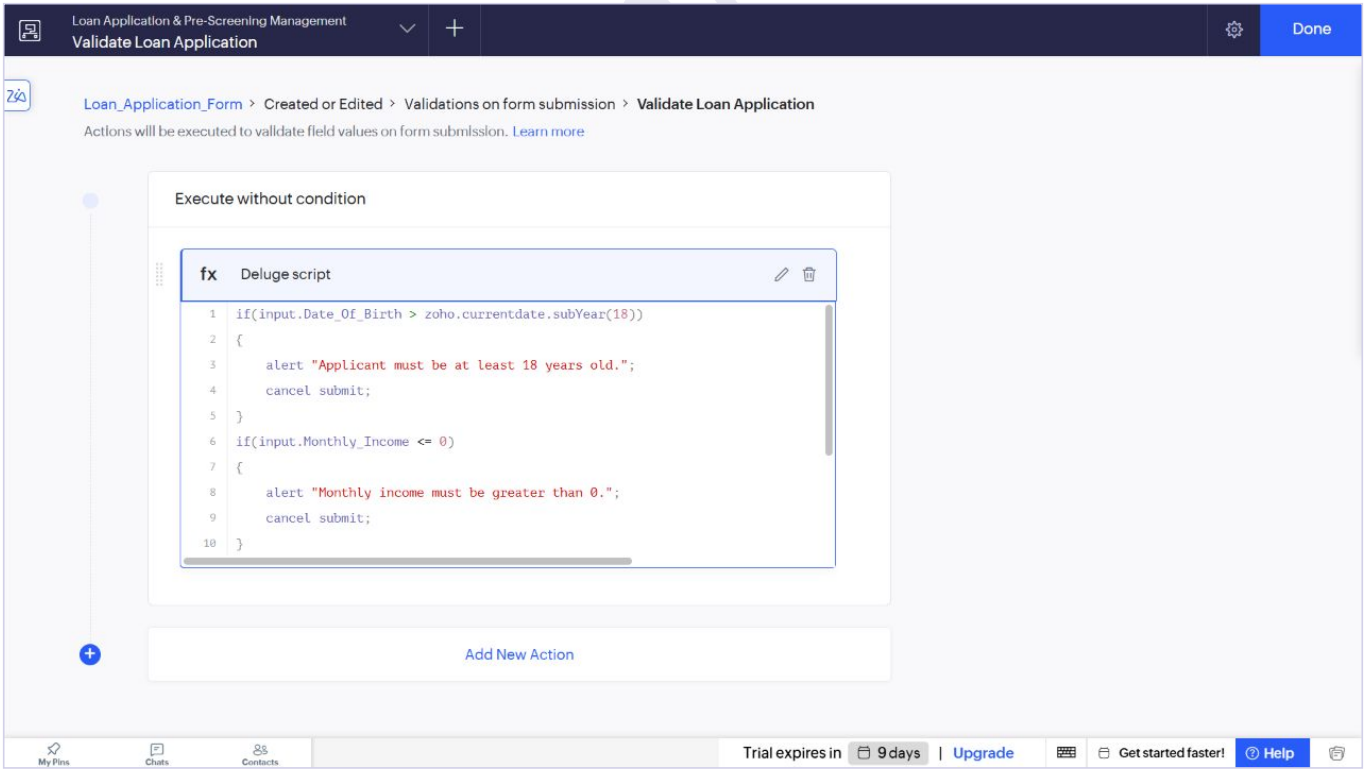


Figure 6.2 - Saved validation action; the complete logic is transcribed on page 3.

7. Runtime Evidence

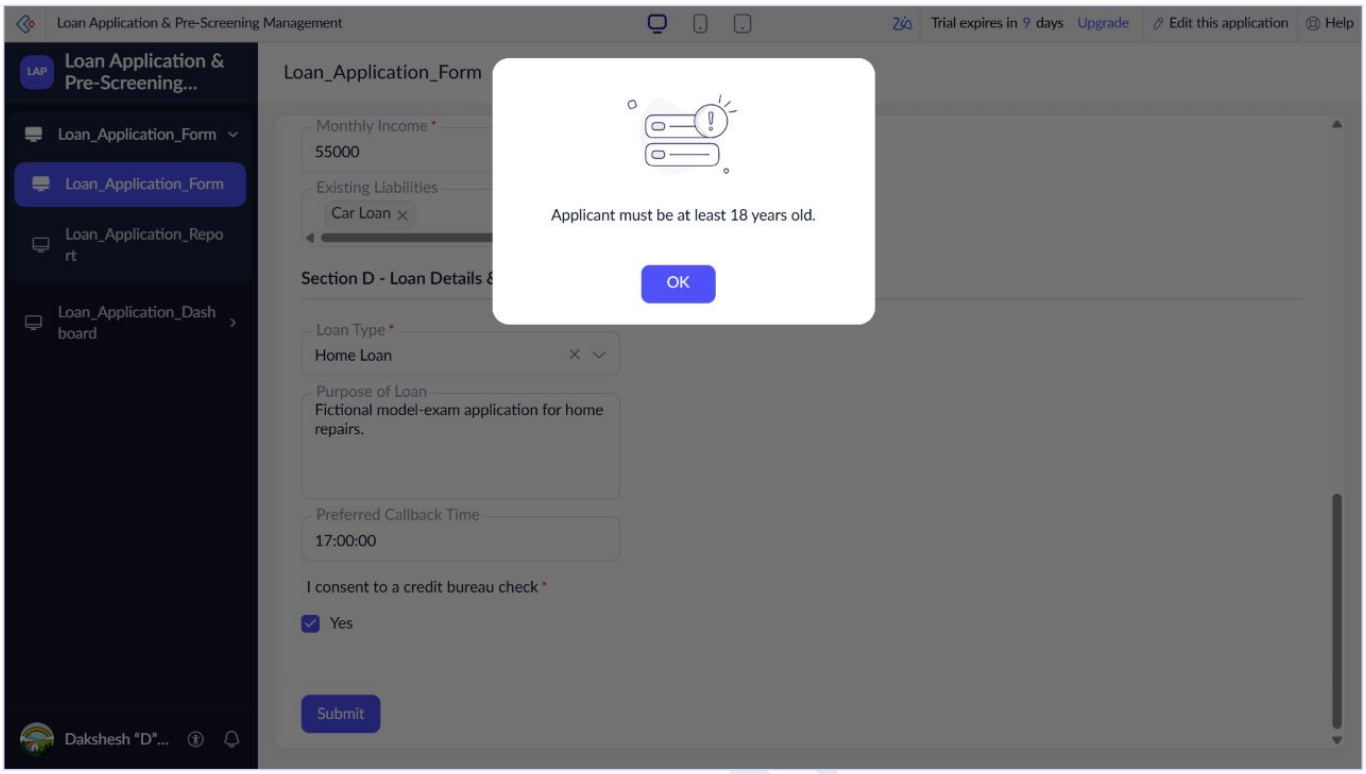


Figure 6.3 - Age validation blocks an under-18 mock application.

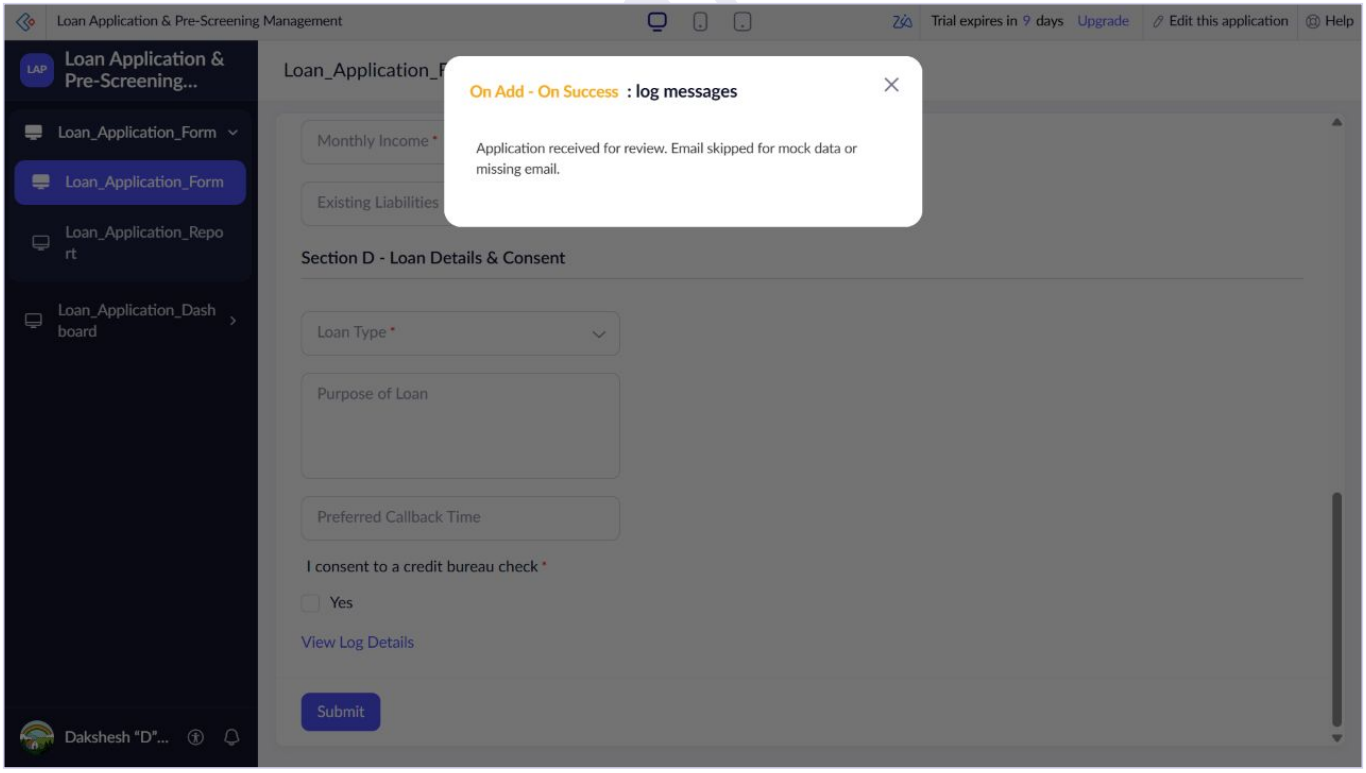


Figure 6.4 - info output confirms the mock acknowledgement path skipped email.

8. Income and Consent Runtime Evidence

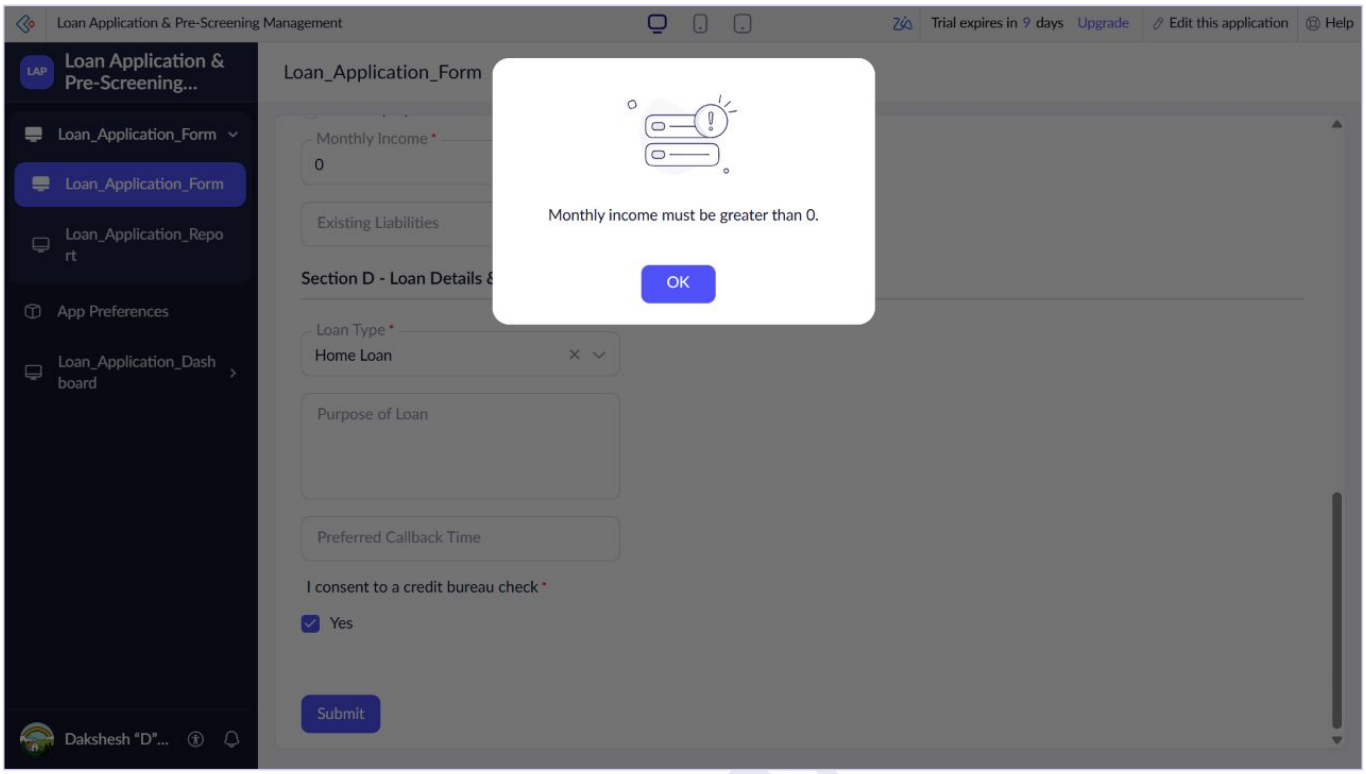


Figure 6.5 - Zero income is blocked before saving the fictional test entry.

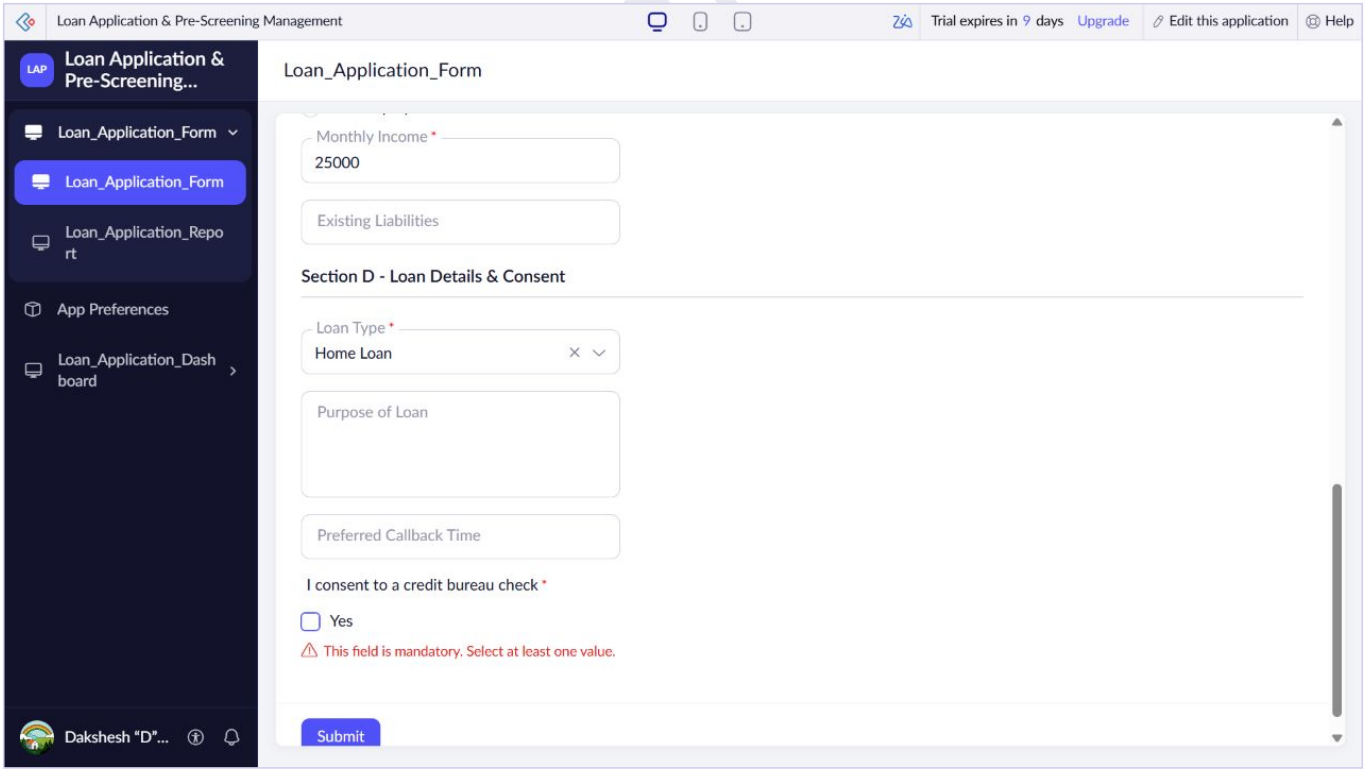


Figure 6.6 - Mandatory consent blocks submission when Yes is unchecked.

9. Test Matrix and Debugging Record

Test	Input / condition	Observed result
T1 - Age	DOB 01-Jan-2015; other required fields supplied	Passed: rejected with age 18+ alert.
T2 - Income	Monthly Income = 0	Passed: positive-income alert; submission rejected.
T3 - Consent	Consent not checked	Passed: mandatory-field rejection.
T4 - Valid mock	Mock Applicant One; income 55,000	Passed: saved; no-email branch used.
T5 - Employment	Mock Applicant Two; Self-Employed; 35,000	Passed: saved.
T6 - Boundary	Mock Applicant Three; DOB 06-Oct-2008 on 06-Oct-2026	Passed: exactly 18; saved.
T7 - Email guard	Missing email or @example.com condition in saved logic	Mock address branch observed in runtime log; no email sent.
T8 - Delivery	Non-mock acknowledgement and admin emails	Not executed; delivery is not claimed.

Debugging Decisions

Validation belongs before saving, so cancel submit is used only in the validation workflow. Acknowledgement belongs after successful submission. The mock guard prevents the classroom examples from attempting delivery. info output gives a simple way to verify the branch taken.

Runtime evidence: screenshots show age rejection, zero-income rejection, mandatory-consent rejection and the mock email-skip message. The two repeated negative tests produced no additional saved application. Live email delivery remains untested.

241401017

10. Result and Learning Outcomes

Basic validation and successful-submission automation are saved and enabled. Negative checks and three valid submissions were tested; the exact age boundary was accepted. Email-delivery behaviour remains a manual follow-up.

Learning outcomes: choose the correct event, reference existing field links, use alert/cancel submit/info, interpret errors and separate syntax checks from runtime and delivery tests.